

Agentic AI for Business and FinTech (FTEC5660)

Homework 02 (Part 1) - Report

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1. System Architecture

1.1 Overview

This CV Verification System is an agentic AI system built to automate the cross-validation of candidate CV information against public Facebook and LinkedIn profiles via the SocialGraph MCP server. The system aims to detect discrepancies between CV content and publicly available online information to assess resume credibility.

1.2 System Architecture

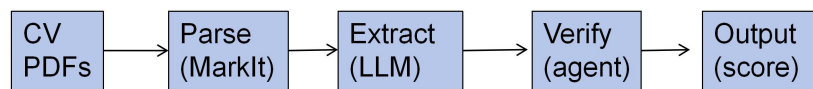


Fig1. System Architecture

1.3 Verify agent details

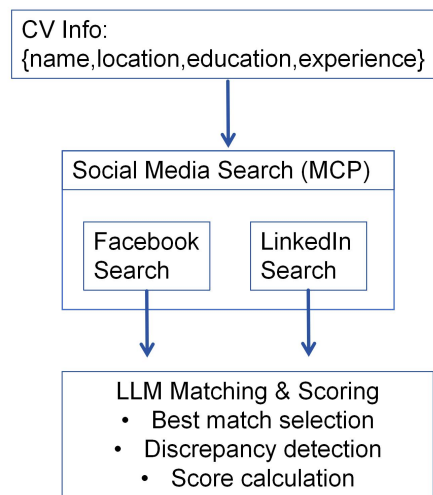


Fig2. Verify Agent Workflow

1.4 Components

1.4.1 Large Language Model Configuration

Component	Specification
Primary Model	DeepSeek Chat (deepseek-chat)

Component	Specification
Temperature	0
Temperature Rationale	Ensures consistent and deterministic responses

Table1. LLM Configuration

1.4.2 Technology Stack

Component	Technology
PDF Parsing	MarkItDown
LLM	DeepSeek Chat
Social Data	MCP Tools (Facebook & LinkedIn)
Framework	LangChain

Table2. Technology Stack

1.4.3 Tool Set

Tool	Function
search_facebook_users	Search users by name
get_facebook_profile	Get Facebook profile
get_facebook_mutual_friends	Get mutual friends
search_linkedin_people	Search people by name
get_linkedin_profile	Get LinkedIn profile
get_linkedin_interactions	Get profile interactions

Table3. MCP Tools List

2. Agent Design

2.1 Agent Workflow

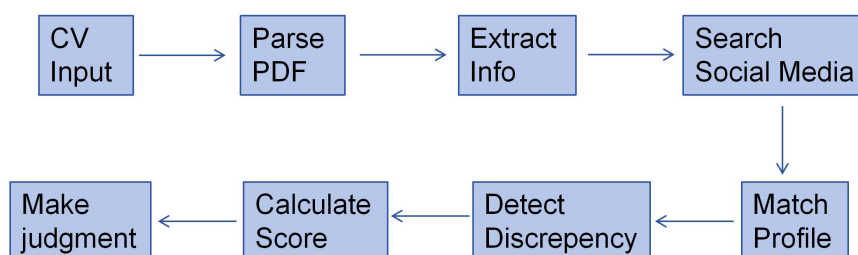


Fig3. Agent Workflow

2.2 Scoring Algorithm

Base Score: 0.35

Bonuses:

- (1) Both FB & LI matched: $+ 0.2 \times \text{FB} + 0.30 \times \text{LI}$
- (2) Only LI matched: $+ 0.3 \times \text{LI}$
- (3) Only FB matched: $+ 0.1 \times \text{FB}$

(4) None: no bonus

Penalties: High -0.2, Medium -0.1, Low -0.05

#Notice: Differences are only checked when a matching experience is found, and penalties are imposed based on these differences.

Judgement:

Score $> 0.5 \rightarrow$ Valid

Score $\leq 0.5 \rightarrow$ Problematic

3. Experiment Result

```
=====
FINAL RESULTS
=====

CV_1.pdf: Score = 0.79
- Decision: Valid CV

CV_2.pdf: Score = 0.64
- Decision: Valid CV

CV_3.pdf: Score = 0.82
- Decision: Valid CV

CV_4.pdf: Score = 0.35
- Decision: Problematic CV

CV_5.pdf: Score = 0.43
- Decision: Problematic CV

=====
Output Format:
=====

scores = [0.7949999999999999, 0.635, 0.825, 0.35, 0.435]

# Ground truth for evaluation (DO NOT MODIFY):
groundtruth = [1, 1, 1, 0, 0]

Evaluation Result: 100.0% correct
```

Fig4. Experiment Result

CV	Name	FB	LI	Score	Decision
CV_1	John Smith	✓	✓	0.79	Valid
CV_2	Minh Pham	✗	✓	0.64	Valid
CV_3	Wei Zhang	✓	✓	0.82	Valid
CV_4	Rahul Sharma	✗	✗	0.35	Problematic
CV_5	Rahul Sharma	✓	✗	0.43	Problematic

Table4. Experiment Result

4. Conclusion

All 5 CVs correctly classified and reach 100% evaluation accuracy.

5. Appendix

Detailed running output:

```
🔥 Starting CV verification process...
=====
📄 Processing: CV_1.pdf
=====
📄 Step 1: Extracting CV information...
Name: John Smith
Location: Singapore, Singapore
Education: 1 entry
Experience: 1 entry

🔍 Step 2: Searching and matching social media profiles...

🔍 Searching for: John Smith
🔍 Found 5 Facebook candidates
🔍 Best FB match: John Smith (confidence: 0.8)
🔍 Found 5 LinkedIn candidates
🔍 Best LI match: John Smith (confidence: 0.95)

🏆 Step 3: Detecting discrepancies and calculating score...

📊 Score: 0.79
🔍 Facebook match: Yes
🔍 LinkedIn match: Yes

✅ No discrepancies found

=====
📄 Processing: CV_2.pdf
=====
📄 Step 1: Extracting CV information...
Name: Minh Pham
Location: Beijing, China
Education: 1 entry
Experience: 2 entrys

🔍 Step 2: Searching and matching social media profiles...

🔍 Searching for: Minh Pham
🔍 Found 5 Facebook candidates
🔍 Found 5 LinkedIn candidates
🔍 Best LI match: Minh Pham (confidence: 0.95)

🏆 Step 3: Detecting discrepancies and calculating score...

📊 Score: 0.64
🔍 Facebook match: No
🔍 LinkedIn match: Yes

✅ No discrepancies found

=====
📄 Processing: CV_3.pdf
=====
📄 Step 1: Extracting CV information...
Name: Wei Zhang
Location: Munich, Germany
Education: 1 entry
Experience: 1 entry

🔍 Step 2: Searching and matching social media profiles...

🔍 Searching for: Wei Zhang
🔍 Found 5 Facebook candidates
🔍 Best FB match: Wei Zhang (confidence: 0.95)
🔍 Found 5 LinkedIn candidates
🔍 Best LI match: Wei Zhang (confidence: 0.95)

🏆 Step 3: Detecting discrepancies and calculating score...

📊 Score: 0.82
🔍 Facebook match: Yes
🔍 LinkedIn match: Yes

✅ No discrepancies found

=====
📄 Processing: CV_4.pdf
=====
📄 Step 1: Extracting CV information...
Name: Rahul Sharma
Location: Singapore
Education: 1 entry
Experience: 2 entrys

🔍 Step 2: Searching and matching social media profiles...

🔍 Searching for: Rahul Sharma
🔍 Found 5 Facebook candidates
🔍 Found 5 LinkedIn candidates

🏆 Step 3: Detecting discrepancies and calculating score...

📊 Score: 0.35
🔍 Facebook match: No
🔍 LinkedIn match: No

✅ No discrepancies found

=====
📄 Processing: CV_5.pdf
=====
📄 Step 1: Extracting CV information...
Name: Rahul Sharma
Location: London | Hong Kong | Singapore
Education: 1 entry
Experience: 4 entrys

🔍 Step 2: Searching and matching social media profiles...

🔍 Searching for: Rahul Sharma
🔍 Found 5 Facebook candidates
🔍 Best FB match: Rahul Sharma (confidence: 0.85)

🏆 Step 3: Detecting discrepancies and calculating score...

📊 Score: 0.43
🔍 Facebook match: Yes
🔍 LinkedIn match: No

✅ No discrepancies found
```